the outer wheels cover larger radius than the inner wheels. The outer wheels rotate more than the inner wheels. Hence the inner wheels have intermittent wear, less grip on the road and withstanding steering issues than outer wheels. The function of the differential unit is to provide different speeds for the inner and outer wheels whenever the vehicle is negotiating a turn.

Construction

Inside the differential casing sun gears are connected with the rear wheel axle. At the left side of the rear axle differential cage is fixed. For connecting two sun gears planets gears are used. At the end of the rear axle road wheels are connected. The construction of differential unit can be seen in Figure 4.9 and the pictorial view of a differential unit can be seen in Figure 4.9 (a).

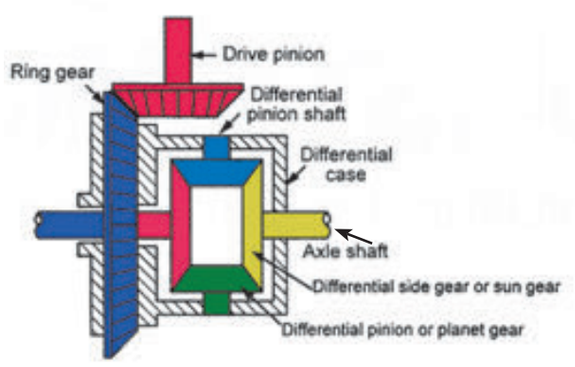


Figure 4.9 Construction of a differential unit

Working principle

When the differential cage rotates, the planet gears inside the differential casing also rotate along with it. Hence due to the rotation of planet gear, the sun gear which is connected to it also rotates and rotates the right side rear axle. Now due to the rotation of the differential cage, sun gears, planet gears, rear axle and rear wheels all rotate at the same speed. When the vehicle takes a turn on the road, the planet gear rotates and tends to rotate the outer wheel at high speed and hence the vehicle turns easily on the road.

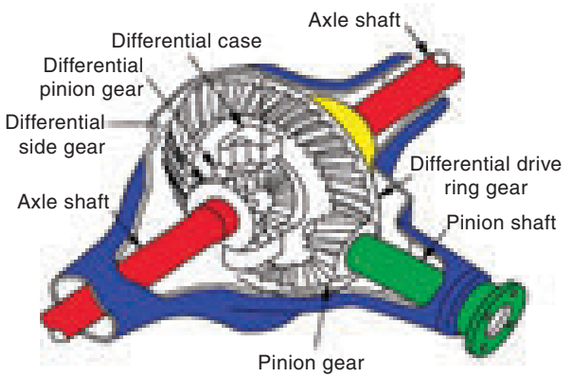


Figure 4.9 (a) Pictorial view of a Differential Unit

When the vehicle goes straight

The power passing from the engine through the clutch, gear box and propeller shaft rotates the pinion shaft of the differential unit. At that time the crown wheel which is connected to the pinion shaft also rotate along with the cage unit. In this condition the planet and sun gears in the differential unit rotate in the opposite direction which creates the brake and the entire unit rotates with full force. Hence the rear axle shaft which is connected to the sun gear rotates at high speed and transmits the power to rear wheels. In this condition the power is transferred at 90° angle and directs the vehicle in a straight direction.

When the vehicle is negotiating a turn

When the vehicle is in a position to take a turn on either right side or

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